

By defn. of free variable each of the n columns has pivots.
 By reasoning from Prob 31, we know that there can be almost n pivots in an $n \times n$ matrix. $\therefore$ There are n pivots and each of the n rows has a pivot.

By Theorem 4 from Chapter 1 and by (a) $\Leftrightarrow$ (d), we can conclude that the equation $Ax=b$ must have a solution for each b in $\mathbb{R}^n$. $\square$

93) $T(x_1, x_2) = (-5x_1 + 9x_2, 4x_1 - 7x_2)$. Show that T is invertible and find a formula for T^{-1} .

Solution:

$$T(x_1, x_2) = \begin{pmatrix} -5x_1 + 9x_2 \\ 4x_1 - 7x_2 \end{pmatrix} = \begin{bmatrix} -5x_1 + 9x_2 \\ 4x_1 - 7x_2 \end{bmatrix} = \begin{bmatrix} -5x_1 \\ 4x_1 \end{bmatrix} + \begin{bmatrix} 9x_2 \\ -7x_2 \end{bmatrix} \quad (\text{Vector addition})$$

$$= x_1 \begin{bmatrix} -5 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 9 \\ -7 \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$= \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$A = \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix} \quad \det A = (-5)(-7) - 9(4) = -1 \neq 0.$$

By Theorem 4, A is invertible. $A^{-1} = \frac{1}{-1} \begin{bmatrix} -7 & -9 \\ -4 & -5 \end{bmatrix}$

$$= \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix}$$

By Theorem 9, T is invertible and $T^{-1} = A^{-1}x$

$$= \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 7x_1 + 9x_2 \\ 4x_1 + 5x_2 \end{bmatrix} \quad (\text{By Row-Vector Rule})$$

95) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible linear transformation. Explain why T is both one-to-one and onto $\mathbb{R}^n$. Use equations (1) and (2). Then give a second explanation using one or more theorems.

Solution: By defn. of invertible, there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $S(T(x)) = x$ for all x in $\mathbb{R}^n$ (1) and $T(S(x)) = x$ for all x in $\mathbb{R}^n$ (2).
 By Theorem 9, A is the standard matrix of T , A is invertible and